

QUITO, Ecuador

WMO: 840710

Lat: 0.141S

Lon: 78.488W

Elev: 9228

StdP: 10.41

Time Zone: -5.00 (W05)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	43.4	45.0	31.8	37.0	59.2	33.9	40.4	58.4	18.9	62.5	17.5	62.6	1.0	340	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.7	71.4	53.3	70.0	53.3	69.4	53.4	58.2	65.8	57.4	65.0	56.7	64.3	8.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.5	93.1	61.0	54.5	89.6	59.6	53.9	87.8	58.8	30.0	66.2	29.4	65.3	28.9	64.7	63.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.3	15.4	13.6	DB	38.9	74.8	1.8	2.1	37.6	76.3	36.6	77.6	35.6	78.8	34.3	80.3	(4)
(5)			WB	35.7	60.3	2.5	0.9	33.8	61.0	32.4	61.5	30.9	62.0	29.1	62.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	57.9	58.1	58.0	58.0	58.1	57.7	58.0	58.4	58.5	58.2	57.7	57.6
	DBStd	2.13	2.21	2.51	2.20	1.79	2.12	1.95	2.27	2.24	2.00	1.96	2.02	1.96
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2550	219	194	218	211	213	218	218	204	195	212	219	229
	CDD50	2926	246	226	248	239	252	232	247	261	255	253	231	236
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	1	0	0	0	0	0	0	0	0	0	1	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.7	4.5	4.4	4.0	3.7	4.0	4.9	5.9	6.3	5.6	4.5	4.1	4.3
Precipitation	PrecAvg	42.6	3.5	4.2	5.3	6.3	4.2	1.9	1.0	1.1	2.9	4.5	4.2	3.3
	PrecMax	57.2	6.7	9.2	11.0	11.5	6.7	4.9	5.7	3.1	5.2	8.8	10.0	4.9
	PrecMin	29.2	0.6	1.1	1.7	3.0	1.5	0.3	0.0	0.1	0.7	0.9	0.4	0.5
	PrecStd	8.5	1.7	2.5	2.0	2.3	1.5	1.1	1.3	1.0	1.3	2.1	2.6	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.3	71.9	71.1	69.9	71.2	70.0	71.1	71.7	71.9	71.8	70.2	69.9
		MCWB	54.5	54.6	54.7	55.2	55.0	53.2	52.1	51.7	51.7	52.9	54.3	54.8
	2%	DB	69.4	69.5	68.3	68.0	68.4	68.3	69.5	69.8	70.1	69.8	68.3	68.1
		MCWB	54.8	55.4	55.0	55.2	54.5	52.7	51.9	51.7	51.8	53.4	54.0	54.8
	5%	DB	67.7	67.6	66.5	66.3	67.7	66.6	67.9	68.2	68.4	68.0	66.6	66.4
		MCWB	54.9	55.0	55.1	55.5	54.4	52.8	51.7	51.6	51.7	53.5	54.3	54.9
	10%	DB	65.8	64.8	64.6	64.5	65.9	65.8	66.2	66.5	66.7	66.1	64.8	64.6
		MCWB	54.8	54.8	54.8	55.2	54.5	52.8	51.8	51.6	52.0	53.8	54.3	54.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.5	58.8	58.5	58.4	58.6	57.1	56.4	56.2	56.6	57.4	57.9	58.5
		MCDB	66.8	66.9	66.0	65.5	66.1	64.7	65.7	66.2	65.9	66.3	65.6	65.9
	2%	WB	57.2	57.4	57.1	57.4	57.2	56.1	54.9	54.7	55.4	56.3	56.7	57.1
		MCDB	65.1	65.1	64.5	64.5	64.5	63.9	64.1	64.7	65.0	65.2	64.6	64.4
	5%	WB	56.3	56.5	56.3	56.6	56.4	54.9	54.0	53.8	54.4	55.3	56.0	56.3
		MCDB	64.0	63.9	63.5	63.5	63.6	62.8	62.9	63.5	63.5	64.0	63.9	63.4
	10%	WB	55.4	55.6	55.4	55.8	55.5	54.1	53.1	52.9	53.5	54.4	54.9	55.2
		MCDB	62.8	62.7	62.4	62.5	62.5	61.9	61.8	62.4	62.0	62.5	62.3	62.1
Mean Daily Temperature Range	5% DB	MDBR	15.9	15.1	15.2	15.0	16.1	17.0	18.1	18.7	19.2	18.3	16.9	16.2
		MCDBR	19.5	18.3	17.9	17.4	19.0	18.9	19.8	20.2	21.2	21.3	19.4	18.8
		MCWBR	8.9	8.3	8.2	8.4	9.0	9.0	9.1	9.2	9.5	9.5	9.2	8.8
	5% WB	MCDBR	16.4	15.9	16.0	15.6	16.6	17.5	18.9	19.7	19.7	18.6	16.6	16.3
		MCWBR	8.0	7.8	7.9	8.1	8.4	8.9	9.1	9.4	9.2	8.8	8.1	8.1
Clear-Sky Solar Irradiance	taub	0.327	0.354	0.346	0.335	0.325	0.304	0.303	0.335	0.353	0.359	0.352	0.342	
	taud	2.534	2.433	2.476	2.519	2.547	2.621	2.616	2.474	2.416	2.417	2.446	2.485	
	Ebn at Noon	318	310	309	306	302	304	306	302	303	304	308	312	
	Edh at Noon	34	38	37	34	32	29	30	35	38	39	37	36	
All-Sky Solar Radiation	RadAvg	1178	1116	1102	1147	1131	1151	1218	1287	1289	1269	1230	1149	
	RadStd	112	89	61	42	70	68	77	99	91	102	115	67	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page